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# HEL-MCNN: Hybrid extreme learning modified convolutional neural network for allocating suitable donors for patients with minimized waiting time

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ABSTRACT

Organ transplantation is the ultimate option to treat terminal illness by transplanting the deceased or damaged organs with healthy organs for improving the patient's life expectancy. The number of organs needed and the organs available for transplantation vary enormously. The tremendous advancements in utilizing big data analytics in the healthcare system make it efficient to explore decision-making information. To make optimal decisions in organ transplantation, this paper proposes a modified convolutional neural network-hybrid extreme learning machine (MCNN-HELM) based prediction model. The proposed MCNN-HELM model utilizes three different real-time datasets as inputs which contain records of liver, heart, and lung transplantation details of the donor and recipient. At first, the missing values and inaccurate data present in real-time datasets are removed via pre-processing. The pre-processed data are then trained using the MCNN-HELM model that efficiently determines the suitable donor for the recipient by minimizing the waiting time of the recipient for the matching organ donor. Moreover, the MCNN-HELM model gives initial preference to patients with high-risk rates to improve their quality of life. The proposed MCNN-HELM model achieves training accuracy of 97.5% with a computational time of 2.2 s, while the precision value of estimated factual outcomes, potential outcomes, and the accuracy of the best donor type are obtained by 16.3582, 16.1401, and 0.6784 which are more efficient than other state-of-the-art methods.

## 1. Introduction

Organ donation is the most demand among countries for the transplantation of organs. It is the process of removing the tissue or organ surgically from one person to the other person. It helps the deceased people to a healthy life and normal life. The important discussion of this organ donation is the advantage of the opt-out system versus the opt-in system for increasing the number of donors. The opt-out systems help in choosing the intention for donating the organ from policymakers. But nowadays, there is only less number of organ donors found among people. The opt-out countries have several kidney and liver transplants (Arshad, Anderson, & Sharif, 2019). The bereaved persons are not ready to give organs of the deceased children (Mahat-Shamir, Hamama-Raz, & Leichtentritt, 2019). Blood donation is a good one used for the welfare of the people which doesn't hurt the human. But, the blood donor may be recovered the within some weeks. The blood print service is managed by the National Transfusion Service and it is also known as the central blood bank. Blood bank supplies blood and blood-related products to the private sector and government hospitals. There is a web application like "Life Share" supports that check the availability of blood requirement. Social media is also helping the requests for organs for a donation of blood (Wijayathilaka, Gamage, De Silva, Athukorala, Kahandawaarachchi, & Pulasinghe, 2020). Organ failure is the most important diagnosed problem in the US. Two difficulties have been identified in increasing organ donation rates. The first is a family agreement for organ donation and willingness to register for organ donation (Khan & Tutun, 2021). The lack of deceased organs is for transplantation and leads to a resurgence of interest in donation after circulatory death (DCD) (Smith, Dominguez-Gil, Greer, Manara, & Souter, 2019). Liver transplantation (LT) is done when the liver is diseased in the end stage. Lack of liver transplantation becomes a challenge; patient survival and graft survival are important research problems. Model for End-Stage Liver Disease (MELD) helps to identify whether this model predicted after LT. The analysis of LT survival is based on pre-LT variables which are profitable

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for liver allocation policies. The improvement of post-LT management lengthens the survival time.

Organ transplantation using machine learning techniques is being conducted in LT. The machine learning techniques have predicted the graft and survival rate accurately when compared with other Cox regression and validated scores (Kazemi, Kazemi, Sami, & Sharifian, 2019). Kidney transplantation helps in curing patients of end-stage renal disease (ESRD). Here, immune suppression impairs host immunity which leads to an increase in the infection risk. The deficiency of limited sampling strategies and high inter and intra-patient variability is used for monitoring and provides direct information on the response of the patient. The machine learning technique helped in immune monitoring is challenging and it helps in understanding immune system complexity (Peng et al., 2020). The manifestation of Delayed graft position (DGF) of renal allograft injury is a common complication seen after the transplantation of deceased donor kidney transplantation (DDKT). The other models have approached Machine Learning (ML) algorithm which learns the patterns from data with programming with rules. Artificial Neural Networks (ANN) and Random Forests (RF) have demonstrated promising results (Miller, Tumin, Cooper, Hayes, & Tobias, 2019). ML models are capable to discriminate between low and high-risk patients 67%-78% of the time which indicates that survival has been predicted. The data accurately and precisely change with excellent results for waitlisted patients (Doby et al., 2021; Kawakita, Beaumont, Jucaud, & Everly, 2020). This paper presents an optimal transplantation system to select compatible donor-recipient with minimized waiting time to enhance a patient's lifespan. At the end of the organ transplant surgery, the patient's life can be saved. Limiting factors include a lack of donors and organs. Sometimes, the right donor cannot be found at the right time (Medved et al., 2018). Improved prediction of outcomes will increase confidence in post-transplant performance and enable optimization of organ allocation. In the initial stage, server locations of the temporary and permanent organ donor collection centers are identified. Since the location of temporary organ collection centers varies, only one server is allocated for the services, and different services are allocated to the permanent collection centers. The existing solutions designed for organ transplant problems were not very cost-effective when it comes to optimal transplant organization systems and minimization of transport time. Most of the authors focused on minimizing the time associated with organ transplantation and the spatial distribution of transplant centers. Others focused on optimizing the time between donor availability and organ transplantation in the recipient. In organ transplantation, the important components are waiting list time, age, region, and the importance of the patient's need.

The organ transplantation cost is mainly associated with the surgery, disposal, hospital, and transplant costs. Organ transplantation centers should be capable of minimizing the waiting time and convincing donors. Hence, the proposed model is designed to overcome the above drawback by allocating the hospital resources to minimize the donors waiting time in organ collection centers and improve the system utility of the recipient transplant patients. To minimize the time of the organ recipients, the service rate and number of servers used are also taken into account. The organ taken from a donor's body is fresh only for a limited amount of time called cold ischemia time. Hence, the transport time serves as a crucial one in the proposed methodology along with the hospital and transplant center allocation time. Furthermore, it will help practitioners more accurately determine the risk of early and late graft failure and improve donor and recipient management. For this reason, the MCNN-HELM method is proposed to find suitable donors with a minimum waiting time. The major contributions of this paper are delineated as follows:

- A novel effective prediction 'modified convolutional neural network-hybrid extreme learning machine (MCNN-HELM)' is proposed to determine a suitable donor for the recipient with minimized waiting time. In addition, it provides an initial preference for patients with high risk based on allocation model policies.
- A prairie dog optimizer (PDO) is utilized to predict the matching pair performance, this algorithm not only reduces the loss rate but also increases the convergence speed of the prediction model.
- The MCNN helps to improve the feature ability of CNN, while HELM is designed to replace the Softmax classifier, thereby further enhancing the performance of transplant systems.
- The performance of the MCNN-HELM model is evaluated using different measures such as precision of estimated factual outcome ($P_{EFO}$), the precision of estimated potential outcome ($P_{EPO}$), the accuracy of best donor type ($A_{BDT}$), training accuracy, and computational time.
- MCNN-HELM model accurately detects suitable donors for the recipient with less waiting time and offers priority for the patients with a high risk of survival for transplanting organs.

The remaining part of the article is portrayed as follows. In section 2, the existing works based on organ donation using machine learning are discussed. The problem formulation is illustrated in section 3 and the proposed approach to organ donation is explained in section 4. The experimental results are presented in section 5. Finally, in section 6, the conclusion of the article is discussed.

## 2. Review of related works

Organ transplantation can be defined as the last resort to treat terminal diseases. Nowadays, transplant waiting lists are a challenging task due to the scarcity of organs and the difficulty of finding donor-recipient compatibility. To overcome the issues a proposed MCNN-HELM is utilized to reduce the waiting time of the patients and also allocate suitable donors.

### A. Literature survey

Various researchers have conducted research for allocating a suitable donor. Geneugelijk and Spierings (2020) developed a concept of organ transplantation using a predicted indirectly recognizable human leukocyte antigen Epitopes (PIRCHE- II) algorithm. Alloreactivity may occur after solid organ transplantation due to a mismatch between contributors and beneficiaries. Salimian and Mousavi (2022) explained the organ donation network design problem concerning climate change and organ quality using robust possibilistic optimization. Multi-objective mixed-integer non-linear programming (MINLP) model is utilized to enhance the quality of organs. Atallah, Badawy, El-Sayed, and Ghoneim (2019) illustrated kidney transplantation outcome prediction using a K-nearest neighbor classifier. Information gain feature selector based Naïve Bayes (IGPFS-NB) based feature selector algorithm; hence it is known as IGBFS-NBBFS/KNN method is used to predict the organ transplantation. Erazo, Goldsman, and Keskinocak (2022) established a simulation-optimization-based framework for organ donation. Rouhani, Pishvaee, and Zarrinpoor (2021) reviewed a fuzzy optimization (FO) model for the transplantation of the organ networks design problem. Berrevoets et al. (2021) explained the allocation of organ transplantation using a queuing-theoretic framework. The organ sync relies on two assumptions namely overlap and unconfoundedness, which fail to predict correctly counterfactual scenarios. Xu et al. (2021) illustrated an organ transplantation allocation to formulate the problem of learning matching representation. Khan, et al. (2021) elaborated on a prediction of organ transplantation using Network-based predictive analytics. A Support Vector Machine (SVM), Logistic Regression (LR), Decision Tree (DT), and Naive Bayes (NB) methods are used to detect organ detection thereby saving more patient lives. Nurdin, Hidayat, and Irvanizam (2022) also analyze Machine learning techniques for the detection and classification of Pneumonia. The CNN-Extreme Gradient Boosting (XGBoost) and CNN-Light Gradient Boosting (LightGBM) are taken for the analysis. Fang (2021) developed a physics-informed neural network (PINN) for partial differential equations (PDEs). Yaseliani, Hamadani, Maghsoodi, and Mosavi (2022) investigate a VGG-based CNN model for the detection of pneumonia in CXR images. Huang et al. (2022) utilized deep cascaded residual networks (DCRNs) for the popular encoder and decoder networks. It can be conducted with three clinical CT datasets. The noise reduction and detail preservation capabilities of the methods are effective with different radiation dose-reduction strategies. Sarvamangala and Kulkarni (2022) reviewed CNN image techniques for the detection of anomalies in the human body.

In conclusion, the literature review shows that many researchers have proposed various deep learning-based models for organ transplantation. The proposed model should design and make optimal decisions in organ transplantation. The proposed model has been evaluated on various real-time datasets as it contains donor and recipient liver, heart, and lung transplantation details. Overall, the literature review highlights the ability of deep learning-based models for organ transplantation. The proposed models have shown predictive results, and further research in this area will lead to the development of more accurate and efficient models for organ transplantation.

B. Limitations of existing methods and research gaps

The organ transplantation models have certain limitations and challenges. Various techniques like PIRCHE- II, MINLP, IGBFS-NB, FO, and SVM have also been developed for organ transplantation. But the existing methods have some limitations such as not being able to better assess the immune risk in solid organ transplantation recipients, and the maximum time and high network cost. Sometimes, it has a low accuracy value and with maximum waiting time, and cannot increase the organ allocation to accurately test new policies. Also, the issues of donors and recipients should be matched incorrectly. MCNN-HELM model is proposed to deal with the above problems. This proposed model can improve the matching donor performance using a prairie dog optimizer and also increase the identification speed using the ELM classifier. Therefore, the proposed approach allocates suitable donors well with minimal waiting time.

## 3. Problem formulation of transplant system

A transplant system includes three components that evolve and interact in real-time. The space of all possible patients is denoted by $\zeta \subset \wp^{e}$, the space of all possible organs represented $\lambda\!\!\!\!- \subset \wp^{f}$. The feature vector of the patient is denoted by $Y \in \zeta$. The feature vector of the organ is represented by $\mathrm{P} \in \lambda\!\!\!\!-$. The observational dataset contains M patients $Y_1, \ldots\ldots, Y_M$. The patients either received or did not receive the organ $\mathrm{P}_j$. Every patient had integrated the survival time $Z_j + X_j$ which breaks the waiting time $X_j \in \wp_+$ denoting the survival post-transplant $Z_j \in \wp_+$ and the time will be spent by the waitlist. The censoring is happen during $Z_j$ and during $X_j$. The objective is performing the interference with the live setting through the streams of organs and patients (Berrevoets et al., 2021). The allocation policy is expressed in the Eq. (1);

$$\pi(u) = \pi(\mathrm{P}(u), R(u)) = k \in \{1, \ldots\ldots, m(u)\} \tag{1}$$

The population life-years are expressed in Eq. (2);

$$\underset{\pi}{MAX} \ \underset{u\to\infty}{LIM} \ \frac{1}{|p(u)|} \sum_{Y \in p(u)} H_{\pi,\lambda_{(u)}}[X + Z|Y] \tag{2}$$

From Eq. (2), $\zeta(u) = \cup_{t\leq u} R(t)$ and the $\lambda\!\!\!\!-(u) = \{\mathrm{P}(u)\}$. Some of the time steps are given and they are;

(1) Better computation of survival with organ, $\mathrm{H}[z(\mathrm{P}(u))|Y]$ for each $Y \in R(u)$. To acquire this computation utilizing censored data and designing $Z(\mathrm{P})$ with the survival analysis through the survival function of the patient's condition is shown in Eq. (3).

$$T(z|Y,\mathrm{P}) = q(Z(\mathrm{P}) > z|Y), \ \forall z \in \wp_+ \tag{3}$$

(2) Better computation of survival without organ, for every $Y \in R(u)$. The survival analyses are utilized for leveraging the censored database to calculate the expected survival without organs.

(3) Approximation of arrival distribution with future organs: The problems are reduced to the system for acquiring the approximation through the leverage of Little's law and multiple queues.

## 4. Proposed methodology of optimal transplantation system

Organ transplantation has become one of the most optimistic advancements in the healthcare system, as which it offers a feasible solution for treating patients with end-stage illnesses to enhance their survival time. The complexities associated with assessing appropriate donor-recipient compatibility and organ scarcity made the transplantation task an arduous process. To overcome this situation, this paper designed an effective predictive model to identify needy recipients by using geo-location information. The organ donation process commences by signing a smart contract to donate organs and by filing an appeal for transplantation. Here, three different real-time datasets containing the records of liver, heart, and lung transplantation data are utilized for evaluation. The incorrect records and missing values present in real-time datasets are removed using the preprocessing step. The preprocessed data is then fed into the proposed optimal transplantation system. The healthcare practitioner analyzes the medical records of the recipient and the smart contract of the donor and decides on matching pair utilizing a modified convolutional neural network- hybrid extreme learning machine (MCNN-HELM) model. The donor and recipient features extracted by a modified convolutional neural network are optimized using a prairie dog optimization algorithm to select a suitable and optimal donor for the recipient. Also, this technique provides an initial preference for patients with a high-risk rate of survival. To minimize the waiting time of the recipient by increasing the donor identification process, the ELM classifier is utilized as the output layer. Fig. 1 shows the overall architecture of the optimal transplantation system.

### 4.1. Data preprocessing

Because of a greater number of incorrect values and missing values, some records and attributes are eliminated. All of the variables of survey administration are eliminated excluding region attributes, as well as the objective of this article, is to investigate the predictive capability of the family consent attributes. One of the important steps of data preprocessing is to remove incorrect records and missing values. The performance of the design is affected because the missing values will remain on the dataset. The missing values are presented on the input file for the neural network and they are taken as valid input values instead of missing values because the prediction designs cannot be recognized as missing values.

### 4.2. Prediction model

This section provides a comprehensive illustration of the proposed modified convolutional neural network- hybrid extreme learning machine (MCNN-HELM) model. To increase the efficiency of transplant systems, the MCNN-HELM model is proposed here for accurate estimation of potential organ donors. An elaborated description of the MCNN-HELM is presented in the following sub-sections.

#### 4.2.1. Modified convolutional neural network (MCNN)

The deep structure and convolutional computation are combined to form CNN which is composed of 3 parts and they are classifier, feature descriptor, and input layer. The feature descriptor contains three layers

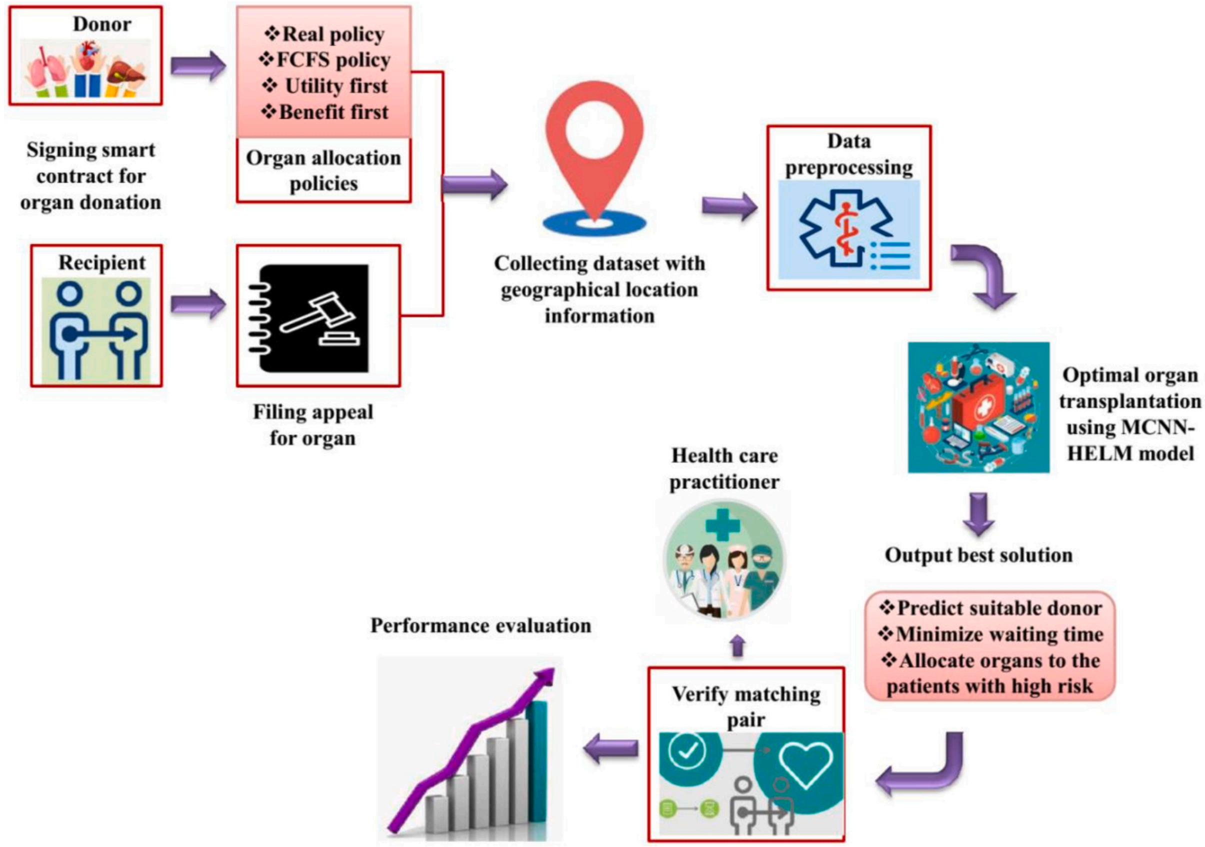


**Fig. 1.** Architecture of optimal transplantation system.

and they are activation layers, multiple convolutional layers, and pooling layers. The classifier consists of fully connected layers like multi-perceptron classifiers for classification and fusion for feature extraction. The CNNs input layer is utilized for preprocessing the multi-dimensional data commonly refers to three-dimensional (3D) data or two-dimensional (2D) data and one-dimensional (1D) data. The convolutional layer comprises multiple convolution kernels for performing the feature learning as well as extracting the input signals. The convolution kernel corresponds to the bias vector and weight matrix and it is equivalent to the neuron of the feed-forward neural networks (Li & Wang, 2020). The convolutional kernels are swept by input features through pre-set stride as well as the activated feature maps based on the receptive field. The convolution process is computed and it is expressed in Eq. (4);

$$Y_j^{(K)} = \sum_{d=1}^{D} X_j^{(d,K)} * Y_{j-1}^{(d)} + C_j^K \tag{4}$$

From Eq. (4), the index of the convolutional layer is represented by $j$, the index of the feature maps in $j^{th}$ layer is denoted by K, $d$ denotes the total number of a convolutional kernel in the $j^{th}$ kernel, the convolution operation is indicated by $*$, then $Y_{j-1}^{(d)}$ represents the input feature of $(j-1)^{th}$ layer, the output feature maps are represented by $Y_j^{(\mathrm{K})}$, the bias and weights of convolution kernels are represented by $C_j^{(\mathrm{K})}$ and $X_j^{(d,\mathrm{K})}$. The non-linear activation function is introduced into network design as well as the feature representation capability will be increased. The tanh, sigmoid, and rectified linear units (ReLU) are the commonly used activation function. One of the important noteworthy functions is ReLU which had effective gradient descent capability as well as avoids disappearance and gradient explosion in the process of training. Then it is expressed in Eq. (5);

$$\begin{cases} \sigma : g = \dfrac{1}{1+e^{-z}} \\ \tanh : g = \dfrac{e^z - e^{-z}}{e^z + e^{-z}} \\ \mathrm{Re}Lu : g = MAX(0, z) \end{cases} \tag{5}$$

The max-pooling process is computed and expressed in Eq. (6);

$$Y_{j+1}^{(\mathrm{K})} = MAX \underset{(j-1)m+1 \le u \le IL}{MAX} Y_j^{(\mathrm{K})}(u) \tag{6}$$

From Eq. (6), after the operation of convolution, the feature map is denoted by $Y_j^{(\mathrm{K})}(u)$, the local area width is represented by $m$, and the output feature map is represented by $Y_{j+1}^{(\mathrm{K})}(j)$. The softmax functions are utilized for computing the probability distributions in the final output layer and it is expressed in Eq. (7);

$$\mathrm{H}_{,\left(y^{(j)}\right)} = \begin{bmatrix} \mathrm{P}\left(z^{(j)} = 1 | y^{(j)};,\right) \\ \mathrm{P}\left(z^{(j)} = 2 | y^{(j)};,\right) \\ \cdots \\ \mathrm{P}\left(z^{(j)} = m | y^{(j)};,\right) \end{bmatrix} = \frac{1}{\sum_{k=1}^{\mathrm{K}} EXP\left(,UY_k^{(j)}\right)} \begin{bmatrix} EXP\left(,UY_1^{(j)}\right) \\ EXP\left(,UY_2^{(j)}\right) \\ \cdots \\ EXP\left(,UY_{\mathrm{K}}^{(j)}\right) \end{bmatrix} \tag{7}$$

From Eq. (7), the total number of classifications is denoted by $n$, the assemblage of arguments of design is represented by $\theta$ as well as the interregional of $z^{(j)}$ is represented by $\{1, 2, \ldots\ldots n\}$. The complex parts of the design are modified or removed by the reduction of trainable metrics with CNN design. The result of feature extraction is a promising solution for enhancing the ability of the design. Particularly, the designs are allowed for handling a smaller quantity of labeled data samples.

In the section, the global average pooling (GAP) is introduced with

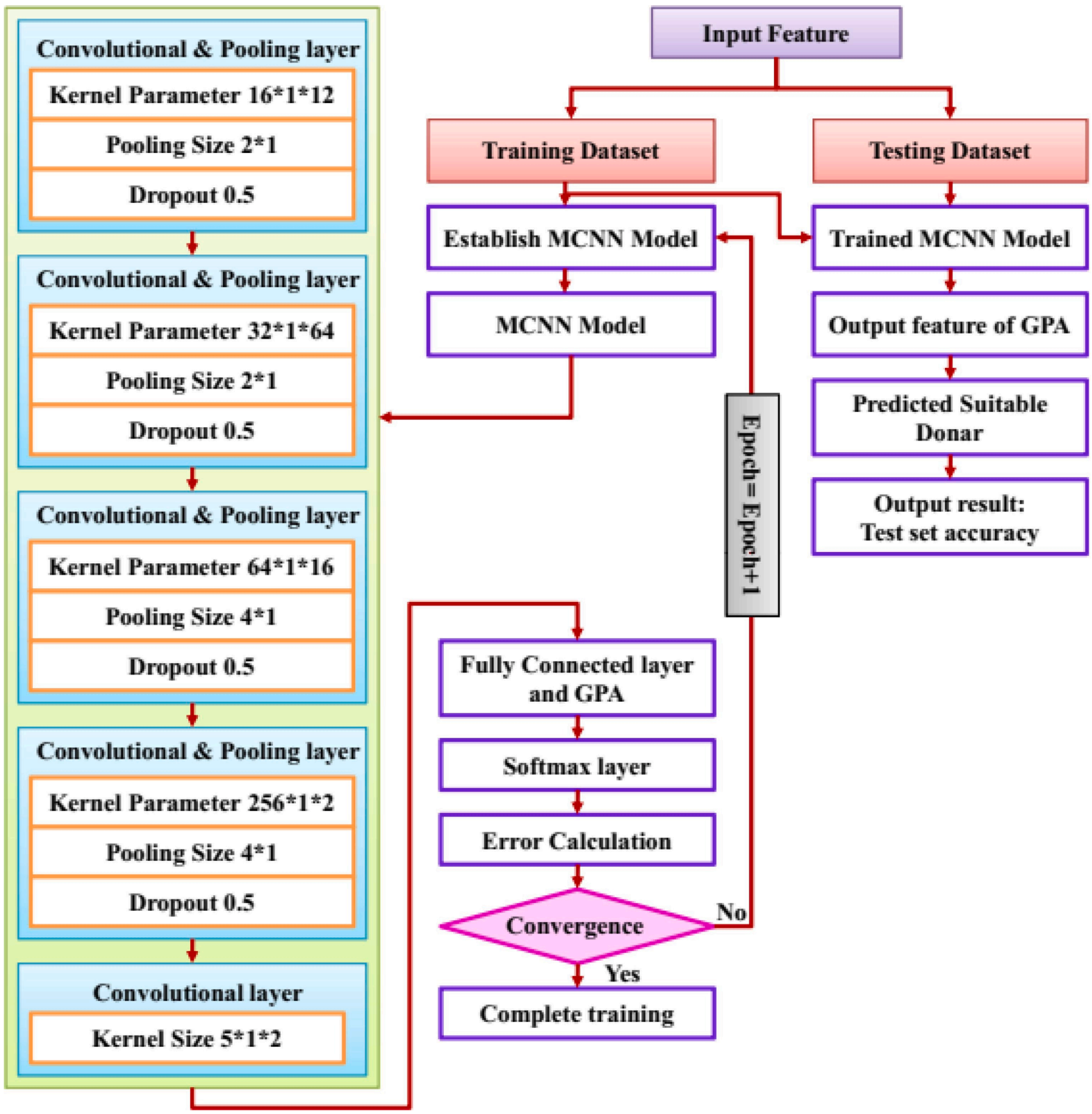


**Fig. 2.** Flowchart of the proposed MCNN model.

dense layers as well as the feature vectors are directly mapped for the output node or category label. This process is known as global average pooling. Reducing the total number of metrics required to train as well as the design is minimized by the training of computational burden. The CNN design is simplified, and then the feature representation capability remains. The convolutional layers, input layers, dropout layers, global average pooling layers, and max-pooling layers are the layers of modified CNN (MCNN). The training model is fed to the predicted outputs and MCNN design. Therefore, the losses among the real and predicted outputs are propagated backwardly for optimizing the network metrics layer through a layer. The optimizer approach is employed for the stochastic gradient descent (SGD) system. The optimization process is calculated and it is expressed in Eq. (8);

$$\theta = \theta - \cdot^{*} \nabla , K\left(\theta; y^{(j)}; z^{(j)}\right) \tag{8}$$

From the above equation, the collection of network metrics is denoted by $\theta$, and the corresponding label and input sample is denoted by $z^{(j)}$ and $y^{(j)}$. The loss function is represented by $K()$, the learning rate is denoted by $\vartheta$, and the gradients are indicated by $\nabla_{\theta}$. Fig. 2 shows the flowchart of the proposed MCNN model and the Pseudocode of the proposed MCNN model is shown in algorithm1.

**Algorithm 1 Pseudocode of MCNN.**

**Algorithm 1:** MCNN
**Input:** Training set, testing set
**Output:**
$C_j^{(K)}$, $X_j^{(d,K)}$: bias and weights of MCNN
**Required parameters:**
*M_ITER*: Maximum number of iteration; $\vartheta$: learning rate; $z^{(j)}$corresponding label; $y^{(j)}$input sample; $K()$: loss function; $\nabla_{\theta}$: gradients
**Initialization works:**
$t$: time, which is initialized as $t = 1$; $L(t)$ is initialized as $L(1) > error$;
**Begin:**
1. Set the required parameters and complete the initialization work
2. **while**$t <$ max_$t$ and $L(t) > error$
3. **for all** training sets:
4. ***rain_p*:** train all the predicted labels
5. **end for**

(*continued on next page*)

(*continued*)

6. Compute the convolution process of output feature maps $Y_j^{(K)}$ using formula (4)
7. Compute the max-pooling process of the output feature map $Y_{j+1}^{(K)}(j)$ using formula (6)
8. Execute the softmax functions of the final output layer $H_{,(y^{(j)})}$ using formula (7)
9. Execute the optimization process $\theta = \theta - \cdot^{*}\nabla, K(\theta; y^{(j)}; z^{(j)})$ using formula (8)
10. Execute the convolution process
11. t++
12. **End while**

**End**

#### 4.2.2. *Extreme learning machine (ELM)*

An ELM is a training algorithm for Single Hidden Layer Feed-Forward Neural Network (SLFN) that gathers much faster than traditional methods and provides optimistic performance. Using the Moore-Penrose pseudo inverse, the bias term and input weights are generated randomly and output weights are determined analytically (Banskota et al., 2022).

Let us consider, N arbitrary samples $(v_x,\ n_x)$, where $v_x = [v_{x1}, v_{x2}, v_{x3}, ..., v_{xt}]^N \in E^t$ and $n_x = [n_{x1}, n_{x2}, n_{x3}, ..., n_{xt}]^N \in E^d (x = 1, 2, ..., T)$. The objective of the regression model is to find the relationship between the target $n_x$ and input $v_x$. ELM is comparable to solving the least square problem thus both bias term and input weights are not needed to be tuned. Mapping the inputs with ELM's hidden neuron and random feature space is the first step.

$$m : v_x \rightarrow m(v_x) \tag{9}$$

From Eq. (9), input weights initially are represented as $m(v_x) = m(u_x.v_y + a_x) \in E^{1\times K}$ and $u_x = [u_{x1}, u_{x2}, u_{x3}, ..., v_{xt}]^N \in E^t$. The bias term is represented as $a_x \in E$. A total number of hidden neurons are indicated as K. $u_x.v_y$ is the dot product of $u_x$ and $v_x$. The hidden output neurons are formulated in equation (10):

$$M = \begin{bmatrix} m(u_1.v_1 + a_1) & \cdots & m(u_K.v_1 + a_K) \\ \vdots & \ddots & \vdots \\ m(u_1.v_T + a_1) & \ldots & m(u_K.v_T + a_K) \end{bmatrix}_{T\times K} \tag{10}$$

The ELM's output is represented in equation (11):

$$N = M\alpha \tag{11}$$

From Eq. (11), ELM's output of the weight is denoted as $\alpha \in E^{K\times d}$ which connects the output of the hidden neurons. $N \in E^{T\times d}$ is represented as the target matrix, which is represented in Eq. (12).

$$\alpha = \begin{bmatrix} \alpha_1^N \\ \vdots \\ \alpha_K^N \end{bmatrix} \quad and \quad N = \begin{bmatrix} n_1^N \\ \vdots \\ n_T^N \end{bmatrix} \tag{12}$$

Using the Moore-Penrose Pseudoinverse, the smallest order least-squares solution can be achieved. This scenario is represented in Eq. (13):

$$\alpha = M^{\lambda} N \tag{13}$$

The Eq. (13), $M^{\lambda}$ is the Moore-Penrose Pseudoinverse of M. When $M^N M$ is not a singular represented as $M^{\lambda} = \left(M^N M\right)^{-1} M^N$, If singular represented as $M^{\lambda} = M^N \left(M M^N\right)^{-1}$.

$$\alpha = \left(M^N M + \frac{1}{C}\right)^{-1} M^N N \tag{14}$$

Eq. (14) shows the ELM network, which works properly with sigmoid and various activation functions. Also, Eq. (15) shows the hyperbolic tangent sigmoid $(\tan sig)$, which is the main function of the research work.

$$\tan sig = \frac{2}{\left(1 + e^{(-2*(u_n.v_y + a_n))}\right)} - 1 \tag{15}$$

At last, power-control devices are implemented using 8-bit fixed-point quantization.

#### 4.2.3. *Prairie dog optimization (PDO)*

The process of optimization begins with the varied coterie forage from one source of food to the other (Ezugwu, Agushaka, Abualigah, Mirjalili, & Gandomi, 2022). The certain response to the voice of unique sound enables the exploitation of the proposed algorithm, and it follows the same random location initialization of prairie dogs as other population-based algorithms.

*4.2.3.1. Initialization of PDO.* By using the uniform distribution, every prairie dog location and coterie are assigned the Eqs. (16) and (17).

$$C_{j,k} = V(01) \times (VG_k - MG_k) + MG_k \tag{16}$$

$$P_{j,k} = V(0,1) \times (vc_k - mc_k) + mc_k \tag{17}$$

Eqs. (16) and (17), $VG_k$ and $MG_k$ represents the $k^{th}$ dimension of the upper and lower bounds of optimization problem. $vc_k = \frac{VG_k}{n}$ and $V(0,1)$ represents a random number with a uniform distribution between 0 and 1. $P_{j,k}$ represents the $k^{th}$ dimension of $j^{th}$ a prairie dog in a coteric and $C_{j,k}$ represents the $k^{th}$ dimension of the $j^{th}$ coterie in a colony.

*4.2.3.2. Fitness function evaluation.* By feeding the solution vector into the fitness function, the fitness function value is evaluated for each prairie dog location. The values of the results are saved in the array and explained in the Eq. (18).

$$g(PD) = \begin{bmatrix} g_1\left(\begin{bmatrix} P_{1,1} & P_{1,2} & \cdots & P_{1,e} & P_{1,e} \end{bmatrix}\right) \\ g_2\left(\begin{bmatrix} P_{2,1} & P_{2,2} & \cdots & P_{2,e-1} & P_{2,e} \end{bmatrix}\right) \\ \vdots \quad \vdots \quad \ldots \quad \vdots \\ g_m\left(\begin{bmatrix} P_{m,1} & P_{n,2} & \cdots & P_{m,e-1} & P_{m,e} \end{bmatrix}\right) \end{bmatrix} \tag{18}$$

Every prairie dog's fitness function value denotes the food quality at a particular source, reaction rightly to anti-predation alerts, and building new burrow potentiality. The minimum fitness function is the exact solution to the minimization problem. The other three regarded the burrow building best value that saves from predators.

*4.2.3.3. Exploration phase of PDO.* The burrow building and foraging of the prairie dog are used for exploring optimization problem space. The burrows of the prairie dog are necessary for protection from predators and the environment. There are two strategies used for the exploration: The first strategy is in the exploration phase, ie for coterie members in scouring the ward for new food sources. The movement of the prairie dog for a food source is caught by the Levy flight motion. In the exploration phase of the algorithm, updates to the position for foraging are given in the Eq. (19).

$$P_{j+1,k+1} = Hbest_{j,k} - fDBest_{j,k} \times \gamma - DP \times Levy(m) \forall iter < \frac{Max_{iter}}{4} \tag{19}$$

The second strategy is used for evaluating the digging strength and assessing the food source's quality. Based on digging strength, the new burrows are built. The digging strength is designed to decline when the number of iterations increases. The burrow building's position updating is given in Eq. (20).

$$P_{j+1,k+1} = HBest_{j,k} \times sP \times ET \times Levy(m) \forall \frac{Max_{iter}}{4} \leq iter < \frac{Max_{iter}}{2} \tag{20}$$

The above equation $fDBest_{j,k}$ evaluates the current effect acquired by the best solution, $HBest_{j,k}$ represents the global best-obtained solution, $sP$ represents the position of the random solution, $\gamma$ represents specialized food source alarm fixed at 0.1 kHz, $DP_{j,k}$ represents randomized prairie dogs' cumulative effect in the colony, ET represents digging strength of coterie and $(m)$ represents the Levy distribution for appreciating superior and more efficient exploration of the problem.

*4.2.3.4. Exploitation phase of PDO.* The main aim of exploitation is utilized in PDO for searching the challenging regions during the intensive exploration phase. The adoption of 2 strategies for this phase is given in Eqs. (21) and (22).

$$P_{j+1,k+1} = HBest_{j,k} - fDBest \times \varepsilon - CP_{j,k} \times rand \forall \frac{Max_{iter}}{2} \leq iter < 3\frac{Max_{iter}}{4} \tag{21}$$

$$P_{j+1,k+1} = HBest_{j,k} \times QF \times rand \ \forall 3\frac{Max_{iter}}{4} \leq iter < Max_{iter} \tag{22}$$

The above equation $HBest_{j,k}$ represents the global best = obtained solution, *fDBest* evaluates the current effect acquired the best solution, $\varepsilon$ is the small number represents the food source quality, the cumulative effect of all prairie dogs in the colony is denoted by $CP_{j,k}$ and $QF$ represents the predator effect.

*4.2.3.5. Computational complexity of PDO.* By the process of initialization, solution position updating, and fitness function evaluation, the computational complexity of PDO is influenced. The computational complexity in the updating position process denoted in $P(QEP) = P(nmMax_{iter}) \times P(Obj)$ function evaluation complexity depends on the optimization problem and it is represented as $P(Obj)$. The mathematical expression of PDO is given in Eq. (23).

$$Q(PD) = Q(nmMax_{iter}) \times Q(Obj) \tag{23}$$

Eq. (23) $Q(m)$ represents the computational complexity of initializing $n$ candidate solutions, $Max_{iter}$ represents the iteration number, $n$ represents the number of candidate solutions, and $m$ represents the number of prairie dogs in a coterie.

### 4.2.4. Proposed MCNN-HELM model for optimal organ transplantation

Fig. 3 describes the working procedure of the proposed modified convolutional neural network- hybrid extreme learning machine (MCNN-HELM) model for effective prediction of needy recipients and suitable donors for optimal transplantation with minimized waiting time utilizing geographical location information. The proposed MCNN-HELM model is evaluated using three distinct real-time organ transplantation datasets along with geographical location information. The inaccurate records and missing values present in the datasets are removed utilizing preprocessing task. The data are available in three real-time datasets and are categorized into two components for testing and training with a ratio of 80:20 respectively.

The training data selected for the training process are fed into the

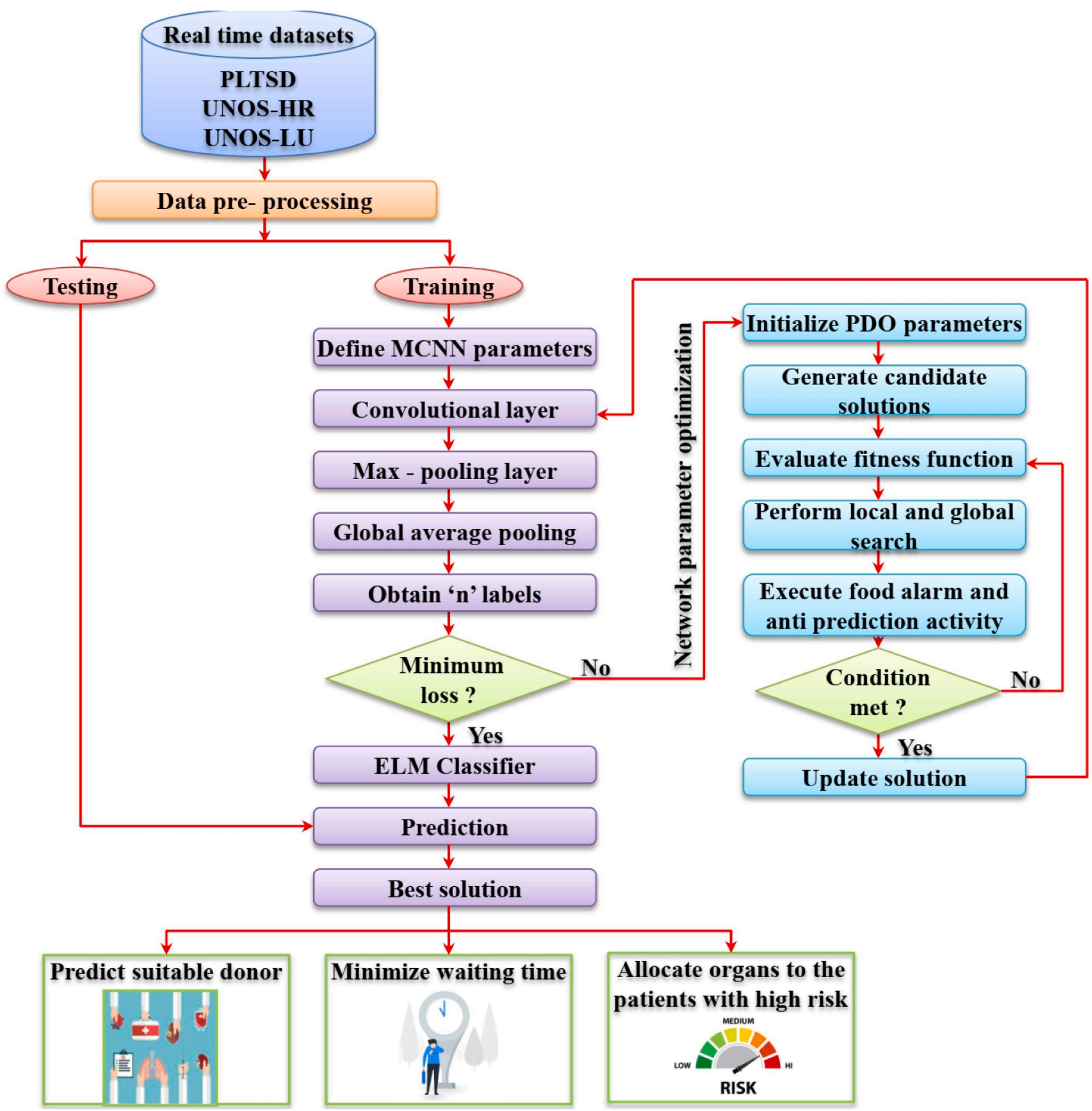


**Fig. 3.** Workflow of proposed MCNN-HELM model.

proposed MCNN-HELM model whereas the testing data are directly evaluated for prediction. Initially, the network parameters of MCNN such as kernel size, pooling size, dropout rate, number of neurons in global average pooling, batch size, and learning rate are initialized randomly. The feature maps generated by convolution and pooling operations are passed through the global average pooling layer that it reduces the number of parameters required and the computational complexity of the training model. Subsequently, the prediction uncertainties between actual and predicted results are estimated. The greater the loss rate, the lesser will be the prediction accuracy. Therefore, the randomly selected parameters are fine-tuned adaptively using the Prairie dog optimization algorithm. This algorithm not only minimizes loss but also maximizes the convergence speed of the prediction model. At each iteration, the features affiliated with high-risk factors are determined and given preference for processing. Taking the advantages such as faster processing time and good generalization ability of ELM classifier into consideration, the softmax classifier is replaced with the ELM classifier. Thus, the proposed MCNN-HELM model accurately detects suitable donors for the recipient with less waiting time and offers priority for the patients with a high risk of survival for transplanting organs.

## 5. Experimental results and discussions

The simulation of the proposed MCNN-HELM model is conducted in the MATLAB platform. For this performance evaluation, three types of datasets such as UK transplant registry data named Paired Liver Transplant Standard Dataset (PLTSD), United Network for Organ Sharing Lung Transplantation (UNOS-LU), and United Network for Organ Sharing Heart Transplantation (UNOS-HR) dataset are used.

### 5.1. Parameter settings of MCNN-HELM

The hyperparameter configuration of the proposed MCNN-HELM model is described in Table 1. The hyperparameter configuration is used to tune the proposed MCNN-HELM model for attaining better performance compared to other state-of-the-art methods.

### 5.2. Evaluation measures

The performance metrics such as training accuracy, computational time, the precision of estimated potential outcomes ($P_{EPO}$), the precision of estimated factual outcomes ($P_{EFO}$), and the accuracy of the best donor type ($A_{BDT}$) are employed for predicting the performance rate. Equation (24), (25), and (26) denotes the precision of estimated outcome, the precision of estimated factual outcomes, and the accuracy of the best donor type. The predicted instances are denoted by True Positive ($T_P$), False Positive ($F_P$), True Negative ($T_N$), and False Negative ($F_N$). Where the correctly predicted positive instances are denoted as $T_P$, the incorrectly predicted positive instances are denoted as $F_P$, the correctly predicted negative instances are denoted as $T_N$, and the incorrectly predicted negative instances are denoted as $F_N$. Accuracy is defined as the percentage of correctly predicted results. The accuracy is calculated by using the equation (27).

$$P_{EPO} = \frac{1}{m}\sum_{j=1}^{m} 1^{T}_{1\times L}\left(\hat{x}^{(j)} - x^{(j)}\right)^2 \tag{24}$$

$$P_{EFO} = \frac{1}{m}\sum_{j=1}^{m} \left(\hat{x}^{(j)}\left[l^{(j)}\right] - x^{(j)}\right)^2 \tag{25}$$

$$A_{BDT} = \frac{1}{m}\sum_{j=1}^{m} \underset{l}{MAX}\,\hat{x}^{(i)}[l] \tag{26}$$

$$Accuracy = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \tag{27}$$

Let us assume $x^{(j)}$ represents the vector outcome under each donor type and $\hat{x}^{(j)}\left[l^{(j)}\right] = \hat{D}\left(y_t^{(j)}, U\left(y_p^{(j)}\right) = l^{(j)}\right)$.

### 5.3. Dataset description

For organ transplantation, the proposed MCNN-HELM model uses three types of real-world datasets. Donor-recipient pairs in the Paired Liver Transplantation Standards Dataset (PLTSD) are extracted from the NHS Liver Transplantation Standards Dataset. This extracted dataset contains 28 donor features and 6,898 pairs of donors, whereas contributors and beneficiaries are classified using 55 features. The lung transplant dataset was obtained from UNOS, including two sub-datasets, 29,210 recipients undergoing lung transplantation and 60,400 recipients undergoing heart transplantation during the period 2012–2022, with 37 contributors and beneficiary characteristics calculated. This real-world data can only be used to monitor transplant outcomes for actual donor-recipient matching, as it does not apply to counterfactual matching. Therefore the simulation is performed for counterfactual matches using transplant outcomes through the semi-synthetic model. The experimentation is conducted for determining the precision of estimated potential outcomes under counterfactual matches and the precision of estimated factual transplant outcomes.

### 5.4. Performance analysis

The number of recipient donor pairs is sampled and various simulations are conducted on the synthetic dataset for relating the proposed MCNN-HELM model's organ allocation policies to the existing model's organ allocation policies. The following organ allocation policies are required for this comparison.

- *Real policy (Real):* According to the expert knowledge, the new donor organ is allotted and the real policy is assumed as an underlying policy because it is generated from the observational data.
- *First Come First Serve (FCFS):* A new contributor is assigned to the first recipient on the queue at FCFS.
- *Utility First (UF):* UF is defined as the new donor organ with the best-predicted transplant outcome assigned to the recipient. On the other hand, survival time after transplantation.
- *Benefit First (BF):* Benefit is the predicted gain in life span after transplantation by assigning a new donor with the highest benefit to the recipient, known as BF.

Here, 80% of the data is selected for training and the remaining 20% of the data is chosen for testing. The simulation results of allocation policies for the PLTSD dataset, UNOS-LU dataset, and UNOS-HR dataset

**Table 1**
Parameter settings of the proposed MCNN-HELM model.

| Methods | Parameters | Ranges |
|---|---|---|
| MCNN | Total number of network layer | 19 |
| | Activation function | ReLU |
| | Kernel size | [64, 128] |
| | Learning rate | 0.001 |
| | Sample size of training set | [128, 256, 512, 1024, 4096] |
| | Sample size of testing set | 800 |
| HELM | Learning rate | 0.01 |
| | Batch size | [5, 10, 20] |
| | Epochs | [8, 16, 32] |
| PDO | Specialized food source alarm | 0.1 kHz |
| | Quality of food source | 2.2204e-16 |
| | PD difference | 0.005 |

**Table 2**
Simulation results of allocation policies for the PLTSD dataset.

| Policies | Death rate | Average benefit | Average survival |
|---|---|---|---|
| Real | 0.23 | 705.98 | 1282.09 |
| FCFS | 0.24 | 703.78 | 1273.28 |
| UF | 0.22 | 725.49 | 1298.35 |
| BF | 0.21 | 739.27 | 1310.56 |

**Table 3**
Simulation results of allocation policies for the UNOS-LU dataset.

| Policies | Death rate | Average benefit | Average survival |
|---|---|---|---|
| Real | 0.15 | 221.48 | 1175.42 |
| FCFS | 0.15 | 215.26 | 1157.38 |
| UF | 0.20 | 110.29 | 1212.92 |
| BF | 0.13 | 275.43 | 1290.48 |

**Table 4**
Simulation results of allocation policies for the UNOS-HR dataset.

| Policies | Death rate | Average benefit | Average survival |
|---|---|---|---|
| Real | 0.15 | 327.31 | 1975.75 |
| FCFS | 0.15 | 325.54 | 1961.74 |
| UF | 0.19 | 206.76 | 2067.58 |
| BF | 0.14 | 379.71 | 2115.23 |

are explained in Table 2, Table 3, and Table 4 respectively.

### 5.5. Comparative analysis

The comparative analysis of the proposed MCNN-HELM model is evaluated by using other state-of-the-art methods such as the MINLP model, IGBFS-NB model, SVM, and Fuzzy Optimization (FO) model. The comparative result of the proposed MCNN-HELM model is explained in Table 5.

The $P_{EFO}$ analysis is performed by using different methods namely MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM which is represented in Fig. 4. The proposed MCNN-HELM model achieved a high precision of estimated factual outcomes of 16.3582 and the MINLP model has a very low precision of estimated factual outcomes of 6.2 compared to other state-of-the-art methods. The remaining models such as IGBFS-NB, SVM, and FO attained the precision of estimated factual outcomes of 8, 11.1, and 14 respectively. Fig. 5 shows the $P_{EPO}$ analysis through various methods such as MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM. The $P_{EPO}$ analysis of 5.8, 9, 11.9, 15, and 16.1401 are obtained from MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM models respectively. From this comparative analysis, the proposed MCNN-HELM model achieved a high precision of estimated potential outcomes and the MINLP model has a very low precision of estimated potential outcomes compared to other methods.

$A_{BDT}$ analysis is conducted using several methods namely MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM which is denoted in Fig. 6. In Accuracy of best donor type analysis, the proposed MCNN-HELM model has a high value of 0.6784 which denoted better performance compared to other state-of-the-art methods. The performances of 0.29, 0.38, 0.46, 0.54, and 0.6784 are obtained from MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM respectively.

Fig. 7 represents the training accuracy of different methods such as MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM. The training accuracy of the proposed MCNN-HELM attained a high-performance rate of 97.5% and the MINLP model has a very low training accuracy of 79%. The remaining methods like IGBFS-NB, SVM, and FO showed a training accuracy rate of 85%, 70%, and 92% respectively. Fig. 8 depicts the computational time of different methods namely MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM. In computational time analysis,

**Table 5**
Comparative results of the proposed MCNN-HELM model.

| Evaluation measures | Ranges |
|---|---|
| $P_{EFO}$ | 16.3582 |
| $P_{EPO}$ | 16.1401 |
| $A_{BDT}$ | 0.6784 |
| Computational time | 2.2 s |
| Training accuracy | 97.5% |

the proposed MCNN-HELM model achieved a very low value of 2.2 s which denoted a better performance compared to other state-of-the-art methods. The computational time of 4.6 s, 4 s, 3.7 s, 3.2 s, and 2.2 s is obtained from MINLP, IGBFS-NB, SVM, FO, and proposed MCNN-HELM respectively.

### 5.6. Computational complexity based on frames and FLOPS

In this section, the MINLP, SVM, FO, and the proposed MCNN-HELM are taken up for a theoretical analysis of the computational complexity, which is measured in terms of floating-point operations (flops). The convolutional function is a very important component of the recent

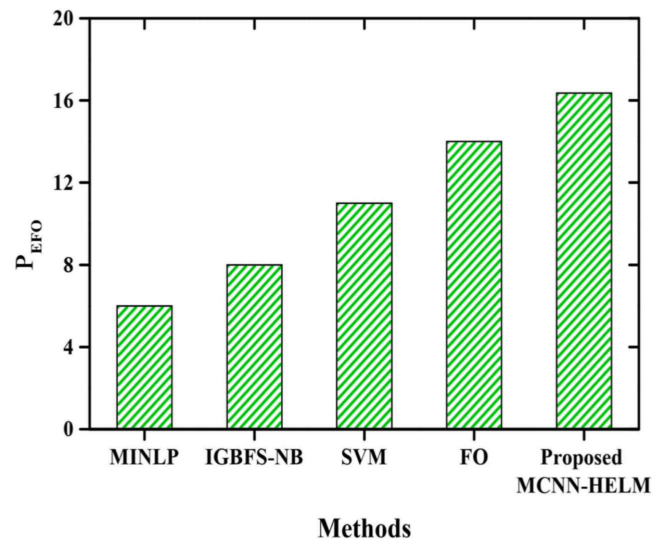


**Fig. 4.** Comparative analysis of.

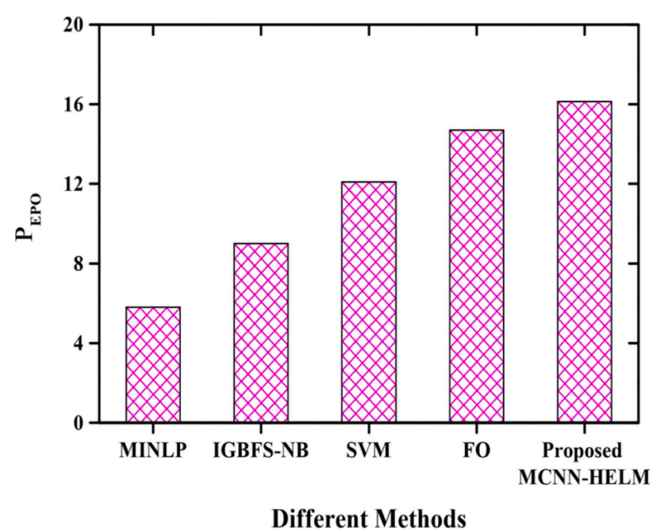


**Fig. 5.** Comparative analysis of.

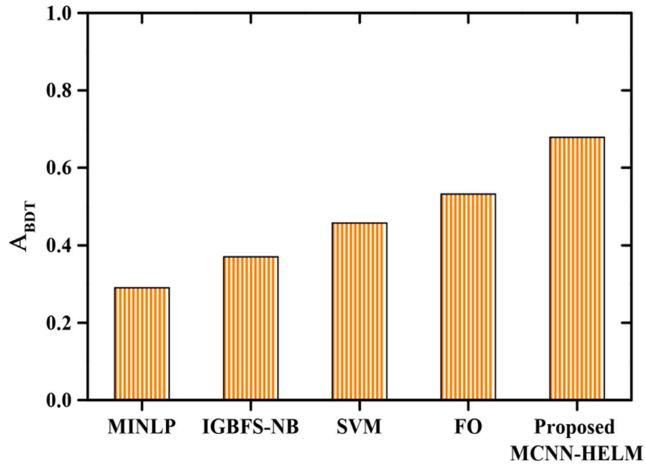


**Fig. 6.** Comparative analysis of.

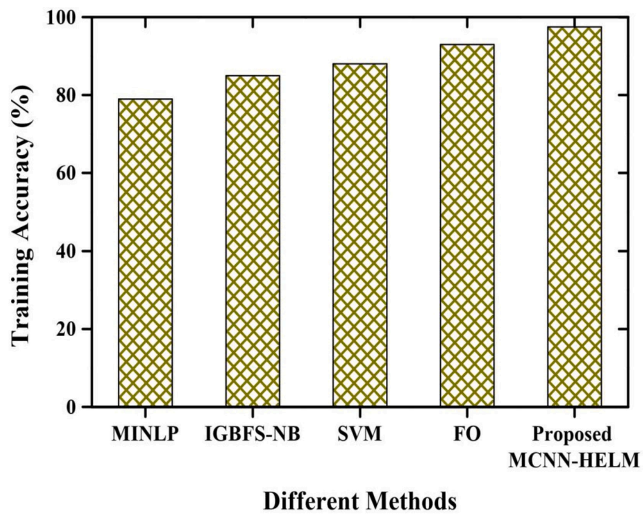


**Fig. 7.** Comparative analysis of training accuracy.

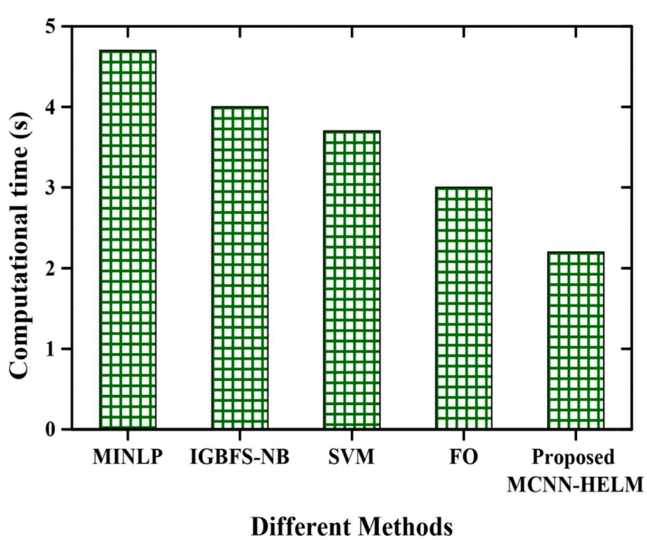


**Fig. 8.** Comparative analysis of computational time.

surge in deep learning research. The computational complexity of MINLP is $\gamma_p^{\delta+1}$ , the value of SVM $\sum_x^p 2^{x^\delta N}$, while the FO is obtained by $2\left(\sum_{x=2}^{p-1} 4p - 4x + 3\right)X - \left(4p^2 - 6p - 4\right)X$. A typical 2D conventional 2D convolution takes $O\left(C_s^2\, K_s^2\, H_{eight}\, W_{idth}\right)$ to calculate. As such computation has become very expensive, these parameters have enhanced over the past few years to meet the needs of demanding applications. The various implementations of convolution have proven to be the most efficient in reducing computational demand. Therefore, the complexity of MCNN-HELM is achieved by$O\left(C_s H_{eight} W_{idth} W_{idth}.\log\left(C_s K_s^2\right)\right)$. Where, $C_s$ denotes the number of channel sizes, $K_s$ is the kernel size, where $H_{eight}$ and $W_{idth}$ are the output width and height respectively. Fig. 9 shows the computational complexity in terms of frames and FLOPs. The frames range from 10 to 100, while the FLOPs range from $10^3$ to $10^9$. The proposed MCNN-HELM obtains the lowest computational complexity among other methods.

## 6. Conclusion

In this section, the MCNN-HELM model is proposed for selecting the donor with minimum waiting time and also providing organ transplantation treatment to high-risk patients. For this performance evaluation, three types of datasets such as PLTSD, UNOS-LU, and UNOS-HR dataset are used. The hyperparameter configuration is used to tune the proposed MCNN-HELM model for attaining better performance compared to other existing techniques. Here, 80% of the data is selected for the training set and the remaining 20% of the data is chosen for the testing set. The performance metrics such as training accuracy, computational time,$P_{EPO}$, $P_{EFO}$, and $A_{BDT}$ are employed for predicting the performance rate. The comparative analysis of the proposed MCNN-HELM model is evaluated by using other state-of-the-art methods such as MINLP, IGBFS-NB, SVM, and FO. In the proposed MCNN-HELM model, the achieved value of precision of estimated factual outcomes is 16.3582, 16.14011 is attained for the precision of estimated potential outcomes, 0.6784 for Accuracy of best donor type, 2.2 s for computational time and 97.5% for training accuracy. The simulation result showed better outcomes compared to other existing methods. In the future, it will concentrate on incorporating additional features linked with family consent. In addition, the proposed system supports approach families by conducting awareness programs to make favorable decisions regarding organ donation to protect one's life.

## CRediT authorship contribution statement

**G. Sangeetha:** Data curation, Writing – original draft, Visualization, Investigation, Software, Validation. **Vanathi Balasubramanian:**

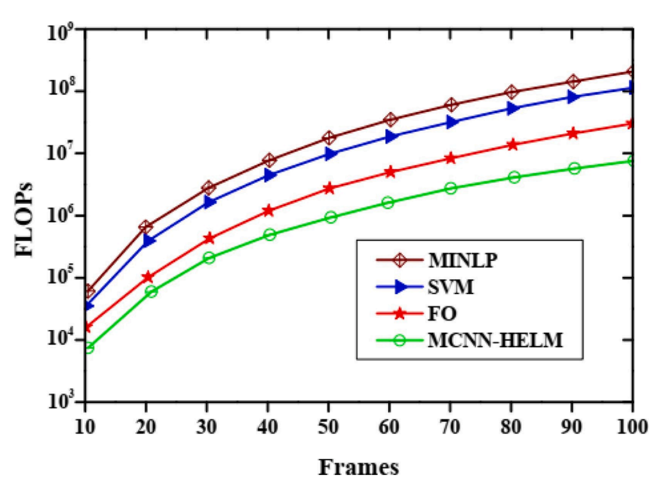


**Fig. 9.** Computational Complexity of FLOPs and Frames.

Conceptualization, Methodology, Software, Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## References

Arshad, A., Anderson, B., & Sharif, A. (2019). Comparison of organ donation and transplantation rates between opt-out and opt-in systems. *Kidney International, 95*(6), 1453–1460.

Atallah, D. M., Badawy, M., El-Sayed, A., & Ghoneim, M. A. (2019). Predicting kidney transplantation outcome based on hybrid feature selection and KNN classifier. *Multimedia Tools and Applications, 78*(14), 20383–20407.

Banskota, N., Alsadoon, A., Prasad, P. W. C., Dawoud, A., Rashid, T. A., & Alsadoon, O. H. (2022). A novel enhanced convolution neural network with extreme learning machine: Facial emotional recognition in psychology practices. *Multimedia Tools and Applications*, 1–25.

Berrevoets, J., Alaa, A., Qian, Z., Jordon, J., Gimson, A. E., & Van Der Schaar, M. (2021). Learning queueing policies for organ transplantation allocation using interpretable counterfactual survival analysis. In *International Conference on Machine Learning* (pp. 792–802). PMLR.

Doby, B. L., Hanner, K., Johnson, S., Purnell, T. S., Shah, M. B., & Lynch, R. J. (2021). Results of a data-driven performance improvement initiative in organ donation. *American Journal of Transplantation, 21*(7), 2555–2562.

Erazo, I., Goldsman, D., & Keskinocak, P. (2022). A Simulation-Optimization Framework To Improve The Organ Transplantation Offering System. arXiv preprint arXiv: 2204.11623.

Ezugwu, A. E., Agushaka, J. O., Abualigah, L., Mirjalili, S., & Gandomi, A. H. (2022). Prairie dog optimization algorithm. *Neural Computing and Applications*, 1–49.

Fang, Z. (2021). A high-efficient hybrid physics-informed neural networks based on convolutional neural network. *IEEE Transactions on Neural Networks and Learning Systems, 33*(10), 5514–5526.

Geneugelijk, K., & Spierings, E. (2020). PIRCHE-II: An algorithm to predict indirectly recognizable HLA epitopes in solid organ transplantation. *Immunogenetics, 72*(1), 119–129.

Huang, Z., Chen, Z., Quan, G., Du, Y., Yang, Y., Liu, X., … Hu, Z. (2022). Deep Cascade Residual Networks (DCRNs): Optimizing an Encoder–Decoder Convolutional Neural Network for Low-Dose CT Imaging. *IEEE Transactions on Radiation and Plasma Medical Sciences, 6*(8), 829–840.

Kawakita, S., Beaumont, J. L., Jucaud, V., & Everly, M. J. (2020). Personalized prediction of delayed graft function for recipients of deceased donor kidney transplants with machine learning. *Scientific reports, 10*(1), 1–13.

Kazemi, A., Kazemi, K., Sami, A., & Sharifian, R. (2019). Identifying factors that affect patient survival after orthotopic liver transplant using machine-learning techniques. *Exp Clin Transplant, 17*(6), 775–783.

Khan, M. E., & Tutun, S. (2021). Understanding and Predicting Organ Donation Outcomes Using Network-based Predictive Analytics. *Procedia Computer Science, 185*, 185–192.

Li, Y., & Wang, K. (2020). Modified convolutional neural network with global average pooling for intelligent fault diagnosis of the industrial gearbox. *Eksploatacja i Niezawodność, 22*(1).

Mahat-Shamir, M., Hamama-Raz, Y., & Leichtentritt, R. D. (2019). Body perception and organ donation: Bereaved parents' perspectives. *Qualitative health research, 29*(11), 1623–1633.

Medved, D., Ohlsson, M., Höglund, P., Andersson, B., Nugues, P., & Nilsson, J. (2018). Improving prediction of heart transplantation outcome using deep learning techniques. *Scientific reports, 8*(1), 3613.

Miller, R., Tumin, D., Cooper, J., Hayes, D., Jr, & Tobias, J. D. (2019). Prediction of mortality following pediatric heart transplant using machine learning algorithms. *Pediatric transplantation, 23*(3), e13360.

Nurdin, Z., Hidayat, T., & Irvanizam, I. (2022). Performance Comparison of Hybrid CNN-XGBoost and CNN-LightGBM Methods in Pneumonia Detection. In *In 2022 International Conference on Electrical Engineering and Informatics (ICELTICs)* (pp. 31–36). IEEE.

Peng, B., Gong, H., Tian, H., Zhuang, Q., Li, J., Cheng, K., & Ming, Y. (2020). The study of the association between immune monitoring and pneumonia in kidney transplant recipients through machine learning models. *Journal of translational medicine, 18*(1), 1–11.

Rouhani, S., Pishvaee, M., & Zarrinpoor, N. (2021). A fuzzy optimization approach to strategic organ transplantation network design problem: A real case study. *Decision Science Letters, 10*(3), 195–216.

Salimian, S., & Mousavi, S. M. (2022). A robust possibilistic optimization model for organ transplantation network design considering climate change and organ quality. *Journal of Ambient Intelligence and Humanized Computing*, 1–24.

Sarvamangala, D. R., & Kulkarni, R. V. (2022). Convolutional neural networks in medical image understanding: A survey. *Evolutionary intelligence, 15*(1), 1–22.

Smith, M., Dominguez-Gil, B., Greer, D. M., Manara, A. R., & Souter, M. J. (2019). Organ donation after circulatory death: Current status and future potential. *Intensive care medicine, 45*(3), 310–321.

Wijayathilaka, P.L., Gamage, P.P., De Silva, K.H.B., Athukorala, A.P.P.S., Kahandawaarachchi, K.A.D.C.P., & Pulasinghe, K.N. (2020 Dec). Secured, intelligent blood and organ donation management system-"LifeShare". In *2020 2nd International Conference on Advancements in Computing (ICAC)* (Vol. 1, pp. 374-379). IEEE.

Xu, C., Alaa, A., Bica, I., Ershoff, B., Cannesson, M., & Van der Schaar, M. (2021). Learning matching representations for individualized organ transplantation allocation. In *International Conference on Artificial Intelligence and Statistics* (pp. 2134–2142). PMLR.

Yaseliani, M., Hamadani, A. Z., Maghsoodi, A. I., & Mosavi, A. (2022). Pneumonia Detection Proposing a Hybrid Deep Convolutional Neural Network Based on Two Parallel Visual Geometry Group Architectures and Machine Learning Classifiers. *IEEE Access, 10*, 62110–62128.